

Linear Diophantine Equations

An equation of the form $a_1x_1 + \cdots + a_nx_n = b$ where $a_1, a_2, \dots, a_n, b$ are fixed integers is called a *linear Diophantine equation*. We assume that $n \geq 1$ and that all coefficients $a_1, \dots, a_n$ are nonzero.

Solvability Condition

The equation $a_1x_1 + \cdots + a_nx_n = b$ is solvable **if and only if** $\gcd(a_1, \dots, a_n) \mid b$.

Structure of Solutions

In case of solvability, all integer solutions can be expressed in terms of $n - 1$ integral parameters.

Proof:

Let $d = \gcd(a_1, \dots, a_n)$. If b is not divisible by d , then the equation is not solvable — for any integers $x_1, \dots, x_n$, the left-hand side is divisible by d while the right-hand side is not.

Proof (contd.)

If $d \mid b$, we obtain the equivalent equation $a'_1x_1 + \cdots + a'_nx_n = b'$, where $a'_i = a_i/d$ and $b' = b/d$, so that $\gcd(a'_1, \dots, a'_n) = 1$. The proof then proceeds by induction on the number n of variables.

The induction proceeds on the number n of variables. For $n = 1$, the equation has the form $x_1 = b$ or $-x_1 = b$, and the unique solution does not depend on any parameter. The key insight is that $\gcd(x_1, \dots, x_n)$ is a *linear combination with integer coefficients* of $x_1, x_2, \dots, x_n$.

- ① For $n = 2$ this follows from Bezout's Lemma. Since $\gcd(x_1, \dots, x_n) = \gcd(\gcd(x_1, \dots, x_{n-1}), x_n)$, we obtain that $\gcd(x_1, \dots, x_n)$ is a linear combination of x_n and $\gcd(x_1, \dots, x_{n-1})$, thus, by induction hypothesis, a linear combination of $x_1, \dots, x_{n-1}, x_n$. $\square$

Corollary

Let a_1, a_2 be relatively prime integers. If (x_1^0, x_2^0) is a solution to $a_1x_1 + a_2x_2 = b$, then **all** solutions are given by:

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General Solution Formula

$$x_1 = x_1^0 + a_2t$$

$$x_2 = x_2^0 - a_1t$$

where $t \in \mathbb{Z}$ is a free integer parameter.

Key Takeaway

Once one particular solution is known, the entire solution set forms a one-parameter family of integer points, spaced evenly along a line in $\mathbb{Z}^2$.

Proof of corollary:

Theorem 1 (Existence Condition). *The linear Diophantine equation $ax + by = c$ has an integer solution if and only if $d \mid c$, where $d = \gcd(a, b)$.*

Proof. We prove this by induction in previous class.

Theorem 2 (General Solution Set). *Let $ax + by = c$ be a linear Diophantine equation where $d = \gcd(a, b)$ and $d \mid c$. If (x_0, y_0) is any particular integer solution to the equation, then the complete set of solutions S is given by:*

$$S = \left\{ \left(x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t \right) : t \in \mathbb{Z} \right\}$$

Proof. To prove that S is the *entire* solution set, we must show two independent propositions:

1. **Correctness:** Every pair $(x, y) \in S$ is a valid solution to the equation.
2. **Completeness:** Every valid solution (x, y) to the equation must belong to S .

The proof of correctness is obvious, so we will do only the proof for completeness.

Let (x, y) be any arbitrary integer solution to $ax + by = c$. We must demonstrate that there exists an integer $t \in \mathbb{Z}$ such that $x = x_0 + \frac{b}{d}t$ and $y = y_0 - \frac{a}{d}t$.

Since both (x, y) and (x_0, y_0) are solutions, we have the following system:

$$ax + by = c \quad (1)$$

$$ax_0 + by_0 = c \quad (2)$$

Subtracting Equation (2) from Equation (1) yields:

$$(ax + by) - (ax_0 + by_0) = c - c$$

Grouping the terms gives:

$$a(x - x_0) + b(y - y_0) = 0$$

which can be rearranged as:

$$a(x - x_0) = -b(y - y_0) \quad (3)$$

Since $d = \gcd(a, b)$, we can divide both sides of Equation (3) by d :

$$\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0) \quad (4)$$

Let us define $a' = \frac{a}{d}$ and $b' = \frac{b}{d}$. Since $d = \gcd(a, b)$, a fundamental property of the greatest common divisor states that the quotients a' and b' are relatively prime; that is:

$$\gcd(a', b') = 1$$

Rewriting Equation (4) using a' and b' , we have:

$$a'(x - x_0) = -b'(y - y_0) \quad (5)$$

From Equation (5), it is clear that b' divides the right-hand side, so b' must also divide the left-hand side:

$$b' \mid a'(x - x_0)$$

Since $\gcd(a', b') = 1$, we can apply **Euclid's Lemma** (which states that if an integer n divides a product kl and $\gcd(n, k) = 1$, then n must divide l). Applying this here, since b' divides $a'(x - x_0)$ and $\gcd(b', a') = 1$, it must be true that:

$$b' \mid (x - x_0)$$

Proof (Contd.)

By definition of divisibility, there exists an integer $t \in \mathbb{Z}$ such that:

$$x - x_0 = b't$$

Substituting $b' = \frac{b}{d}$ back into this expression gives:

$$x - x_0 = \frac{b}{d}t \implies x = x_0 + \frac{b}{d}t$$

This establishes the required form for x . To isolate y , we substitute $x - x_0 = b't$ back into Equation (5):

$$a'(b't) = -b'(y - y_0)$$

Since $b \neq 0$, we have $b' \neq 0$, allowing us to divide both sides by b' :

$$a't = -(y - y_0)$$

Rearranging this equation to isolate y gives:

$$y - y_0 = -a't \implies y = y_0 - a't$$

Substituting $a' = \frac{a}{d}$ back into the equation yields:

$$y = y_0 - \frac{a}{d}t$$

Thus, we have shown that the arbitrary solution (x, y) must be of the form:

$$x = x_0 + \frac{b}{d}t \quad \text{and} \quad y = y_0 - \frac{a}{d}t$$

for some $t \in \mathbb{Z}$. This proves that the set S contains all possible solutions, concluding the proof of completeness. $\square$

Example problems on GCD and Bézout's Lemma:

Example 1:

Prove that the expression

$$\frac{\gcd(m, n)}{n} \binom{n}{m}$$

is an integer for all pairs of integers $n \geq m \geq 1$.

Hint: Use Bézout's Lemma.

Solution:

Solution. By Bezout's identity, there exist integers a and b such $\gcd(m, n) = am + bn$. Next, notice that

$$\frac{\gcd(m, n)}{n} \binom{n}{m} = \frac{am + bn}{n} \binom{n}{m} = \frac{am}{n} \binom{n}{m} + b \binom{n}{m}.$$

We must now prove that $\frac{am}{n} \binom{n}{m}$ is an integer. Note that

$$\frac{m}{n} \binom{n}{m} = \frac{m}{n} \left(\frac{n!}{m!(n-m)!} \right) = \frac{(n-1)!}{(m-1)!(n-m)!} = \binom{n-1}{m-1}.$$

Therefore,

$$\frac{\gcd(m, n)}{n} \binom{n}{m} = a \binom{n-1}{m-1} + b \binom{n}{m},$$

which is clearly an integer. □

Example 2:

[HMMT 2002] Compute

$$\gcd(2002 + 2, 2002^2 + 2, 2002^3 + 2, \dots).$$

Hint: Compute gcd of the first two terms.

Solution:

Solution: Let g denote the desired greatest common divisor. Note that $2002^2 + 2 = 2002(2000 + 2) + 2 = 2000(2002 + 2) + 6$. By the Euclidean algorithm, we have

$$\gcd(2002 + 2, 2002^2 + 2) = \gcd(2004, 6) = 6.$$

Hence $g \mid \gcd(2002 + 2, 2002^2 + 2) = 6$. On the other hand, every number in the sequence $2002 + 2, 2002^2 + 2, \dots$ is divisible by 2. Furthermore, since $2002 = 2001 + 1 = 667 \cdot 3 + 1$, for all positive integers k , $2002^k = 3a_k + 1$ for some integer a_k . Thus $2002^k + 2$ is divisible by 3. Because 2 and 3 are relatively prime, every number in the sequence is divisible by 6. Therefore, $g = 6$. $\square$

Example 3:

Problem 1.3.10. The natural numbers a and b are such that

$$\frac{a+1}{b} + \frac{b+1}{a}$$

is an integer. Show that the greatest common divisor of a and b is not greater than $\sqrt{a+b}$.

(1996 Spanish Mathematical Olympiad)

Hint: Let $g = \gcd(a,b)$. Write $a = g \cdot m$ and $b = g \cdot n$, where $\gcd(m,n) = 1$.

Solution:

Solution. Let $d = \gcd(a, b)$ and put $a = md$ and $b = nd$. Then we have

$$\frac{md + 1}{nd} + \frac{nd + 1}{md} = \frac{m^2d + m + n^2d + n}{mnd}$$

is an integer, so that in particular, d divides $m^2d + m + n^2d + n$ and also $m + n$. However, this means $d \leq m + n$, and so $d \leq \sqrt{d(m + n)} = \sqrt{a + b}$.

Example 4:

Problem 1.3.13. Several positive integers are written on a blackboard. One can erase any two distinct integers and write their greatest common divisor and least common multiple instead. Prove that eventually the numbers will stop changing.

(1996 St. Petersburg City Mathematical Olympiad)

Hint: Consider the sum and product of the numbers on the blackboard.

Solution:

Solution. If a, b are erased and $c < d$ are written instead, we have $c \leq \min(a, b)$ and $d \geq \max(a, b)$; moreover, $ab = cd$. From this we may conclude $a + b \leq c + d$ writing $ab + a^2 = cd + a^2 \leq ac + ad$ (the latter since $(d - a)(c - a) \leq 0$) and dividing both sides by a . Thus the sum of the numbers never decrease, and it is obviously bounded (e.g. by n times the product of the numbers, where n is the number of numbers on the board); hence it eventually stops changing, at which time the numbers never change.